

Human Development Index (HDI) Predictor

Project Documentation

Project: Human Development Index (HDI) Predictor

Phase: Project Documentation

1. Project Summary

The HDI Predictor is a Flask-based machine learning web application that predicts a country's Human Development Index (HDI) score and development category from four indicators: Life Expectancy, Mean Years of Schooling, GNI per Capita, and Internet Users.

2. Architecture Summary

A 3-layer architecture: a User Layer (browser forms), a Flask Application Layer (routes for Home, Prediction, and prediction handling), and a Model Storage & Training Layer (the dataset, EDA, preprocessing, and the trained Linear Regression model).

3. Data Pipeline Summary

- Dataset generation (144 countries) with the official UNDP HDI formula
- EDA — correlation heatmap and strip plot
- Preprocessing — null-value imputation and Country label encoding
- Model training and evaluation ($R^2 \approx 0.99$)
- Model serialization with Pickle

4. Model Performance

Metric	Value
R^2	≈ 0.99
MAE	≈ 0.013
RMSE	≈ 0.014

5. References

- UNDP Human Development Report methodology (HDI formula)
- Scikit-learn documentation — scikit-learn.org
- Flask documentation — flask.palletsprojects.com